

BCSE355L-Cloud Architecture Design-Project
NOTEFY – STUDENT NOTES SHARING AND TASK
MANAGEMENT WEBSITE

Submitted in partial fulfillment of the requirements for the Internals in

Course Project

in

B. Tech [All Branches]

by

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(Deemed to be University under section 3 of UGC Act, 1956)

November 2025

DECLARATION

I hereby declare that the project entitled **NOTEFY – STUDENT NOTES SHARING AND TASK MANAGEMENT WEBSITE**

submitted by me, for the award of the degree of *Bachelor of Technology in Computer Science and Engineering* to VIT is a record of bonafide work carried out by me under the supervision of Prof. / Dr. Yoganand S.

I further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place :

Vellore

Date :

**Signature of the
Candidate**

CERTIFICATE

This is to certify that the project entitled **NOTIFY – STUDENT NOTES SHARING AND TASK MANAGEMENT WEBSITE** submitted by Aarya Nagverkar 23BCI0061 and Medha S 23BAI0039. **School of Computer Science and Engineering**, VIT, for the award of the degree of *Bachelor of Technology in Computer Science and Engineering*, is a record of bonafide work carried out by her under my supervision during Fall Semester 2024-2025, as per the VIT code of academic and research ethics.

The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university. The project fulfills the requirements and regulations of the University and in my opinion meets the necessary standards for submission.

Place : Vellore

Date :

**Signature of
the Guide**

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ABSTRACT

This project titled “NOTEFY – Student Notes Sharing and Task Management Website” is a cloud-based academic collaboration platform developed to simplify resource sharing and personal productivity for students. The system allows authenticated users to upload, access, and share academic materials through an organized digital repository while managing their coursework and deadlines via a task management module. By integrating Amazon Web Services (AWS) cloud infrastructure, the project demonstrates the real-world application of scalable and secure cloud architectures for education-oriented web platforms.

The frontend of NOTEFY is developed using React.js, offering a dynamic and responsive interface consistent with modern UI/UX standards. The backend uses Node.js with Express.js to handle API logic and routing, while AWS S3 manages file storage and retrieval. EC2 instances host the backend, and AWS Cognito is used for secure authentication and session handling. DynamoDB or RDS stores metadata such as user information, file links, and task lists. The architecture follows a three-tier model presentation, application, and data layers ensuring modularity and fault tolerance.

This project underscores the importance of centralized, cloud-hosted note-sharing systems in academia. It bridges the gap between generic storage applications and domain-specific educational collaboration tools. By adopting AWS, NOTEFY achieves cost efficiency, security, and high availability while providing a simple yet powerful platform for students to collaborate, organize, and learn effectively.

CHAPTER 1

1. INTRODUCTION

1.1 Background

Cloud computing has revolutionized how information is stored, processed, and accessed across the world. It enables organizations and individuals to use on-demand computing resources without investing heavily in physical infrastructure. Within the educational sector, this transformation has been particularly significant: students, teachers, and institutions increasingly depend on cloud-based platforms for collaboration, data sharing, and online learning.

Despite this shift, many widely used platforms such as Google Drive, Dropbox, and OneDrive are primarily designed for general storage and lack academic-specific features such as subject-wise categorization, peer validation, and seamless task organization. This leads to scattered study resources and inconsistent collaboration among students.

To bridge this gap, **NOTEFY** introduces a cloud-integrated system that enables structured academic resource sharing alongside personalized task management. By leveraging the scalability and reliability of **Amazon Web Services (AWS)**, the platform ensures secure access, high availability, and efficient management of learning materials within a unified ecosystem.

1.2 Motivation

In modern academic environments, students are often required to handle numerous assignments, projects, and examinations simultaneously. However, their study materials and notes are frequently dispersed across different storage media and communication platforms. This fragmentation not only wastes time but also limits the collaborative potential of peer learning.

The motivation behind developing NOTEFY stems from the need for a **centralized, student-centric platform** that allows learners to easily share, retrieve, and organize study resources while keeping track of individual academic goals.

Moreover, the project aims to demonstrate how **cloud-based infrastructure** can be used to deliver an affordable, scalable, and intuitive application tailored for educational use.

Through an interactive and responsive web interface built with React JS and a Node.js backend, the system encourages digital collaboration and fosters academic productivity.

1.3 Scope of the Project

The scope of NOTEFY extends beyond a conventional note-sharing website. It encompasses the design, development, and deployment of a **full-stack web application** that integrates cloud computing principles and modern web technologies. The project covers:

- Implementation of **secure authentication** and user management through AWS Cognito.
- Cloud-hosted **file storage and retrieval** using AWS S3 buckets.
- Backend deployment on **AWS EC2**, enabling continuous uptime and scalability.
- Integration of a **task management module** allowing students to plan and monitor their academic progress.
- Development of a **responsive frontend** using React JS for an intuitive and seamless user experience.

In addition to functionality, the project explores aspects of system security, cost optimization, and sustainability in a cloud environment. The outcome contributes not only as a student productivity tool but also as a **demonstration of effective AWS-based web application design**, suitable for future extensions such as AI-driven recommendations or real-time collaboration features.

CHAPTER 2

2. PROJECT DESCRIPTION AND GOALS

2.1 Literature Review

With the growing adoption of digital tools in education, several platforms have emerged to support online learning, document sharing, and collaboration. Popular applications such as **Google Classroom**, **Notion**, and **Microsoft OneNote** allow students to share and organize learning materials. However, these tools are often **general-purpose** rather than tailored to academic contexts. For example, Google Classroom focuses on assignment submission and teacher–student workflows, while Notion provides flexible documentation but lacks a structured file repository or integrated task tracker for academic use.

Research in cloud computing for education highlights that while cloud-based solutions increase accessibility and scalability, the lack of customization for academic collaboration remains a limitation. Tools like **Dropbox** or **Google Drive** provide file synchronization but do not support metadata tagging, subject categorization, or peer-contributed content organization.

Studies also emphasize the growing importance of **student-driven content ecosystems**, where learners actively contribute and curate resources. This pedagogical shift calls for platforms that blend collaboration, task tracking, and scalable infrastructure, a niche that NOTEFY directly addresses.

By leveraging **AWS Cloud Services**, this project integrates the scalability of industrial-grade infrastructure with the simplicity required by students. It introduces a dedicated academic ecosystem that merges file storage, sharing, and task organization, something existing systems have yet to achieve comprehensively.

2.2 Research Gap

Although cloud-based educational platforms are widely used, most suffer from one or more of the following limitations:

1. **Lack of academic context:** General-purpose cloud tools do not provide discipline-wise categorization or filtering, making it harder for students to locate relevant material.

2. **Limited collaboration:** Many platforms allow sharing but lack peer validation, rating, or feedback mechanisms that promote academic engagement.
3. **No integrated productivity tracking:** Students must rely on separate apps (e.g., Google Tasks, Trello) to manage their assignments, which fragments the workflow.
4. **Deployment complexity:** Few student-oriented tools demonstrate scalable, real-world deployment using modern cloud technologies.

To fill these gaps, NOTEFY combines the core principles of **cloud computing, web engineering, and human-computer interaction (HCI)** into a single unified platform. It provides:

- Secure, subject-oriented note sharing.
- Seamless file uploads and downloads through **AWS S3**.
- Integrated task management to monitor personal academic progress.
- A fully deployable cloud infrastructure that can be scaled institution-wide.

This research gap positions NOTEFY not only as a student aid but as a **prototype for educational cloud systems** that can scale across universities and communities.

2.3 Project Plan

The project was executed following an **Agile development methodology**, emphasizing iterative design, feedback, and testing. Each sprint was structured to handle specific modules of the system: frontend design, backend development, and AWS integration, ensuring progressive enhancement at every stage.

Phase 1 – Requirement Analysis:

- Defined functional and non-functional requirements.
- Identified AWS services suitable for deployment and storage.

Phase 2 – System Design:

- Designed architecture diagrams outlining communication between frontend, backend, and AWS services.
- Created low-fidelity and high-fidelity UI mock-ups using Figma.

Phase 3 – Implementation:

- Developed frontend using **React.js** for modular component rendering and responsive layouts.
- Implemented RESTful APIs with **Node.js and Express.js**.
- Integrated AWS S3 for secure file storage and retrieval.
- Configured AWS Cognito for authentication and authorization.

Phase 4 – Testing and Debugging:

- Conducted unit and integration testing on all modules.
- Simulated concurrent uploads to evaluate AWS scalability and reliability.

Phase 5 – Deployment and Documentation:

- Deployed backend using **AWS EC2** instances and connected them with S3 and RDS/DynamoDB.
- Documented APIs and workflows for maintainability and further enhancements.

The outcome of this plan is a **fully functional and cloud-deployed academic collaboration system**, designed to be extensible and reusable in future coursework or institutional projects.

CHAPTER 3

3. TECHNICAL SPECIFICATIONS

3.1 Requirements

3.1.1 Functional Requirements

- **User authentication and session management via AWS Cognito**

To ensure data privacy and secure access, the system integrates AWS Cognito for user registration, login, and session handling. Each user is authenticated through a unique identity token, allowing personalized access to uploaded notes and tasks. This guarantees that sensitive user information remains protected under industry-grade security standards.

- **File upload, download, and sharing using AWS S3**

All study materials uploaded by users are stored in Amazon S3 buckets. S3 provides high-durability object storage and makes it easy to manage files through presigned URLs. This enables seamless uploads and downloads while ensuring fast retrieval and reliable backup through AWS's distributed architecture.

- **CRUD operations for task management**

The platform includes a built-in to-do or task tracker where users can create, read, update, and delete tasks. This module allows students to organize their academic schedules efficiently, monitor deadlines, and maintain personal productivity within the same environment used for sharing notes.

- **Subject-wise categorization of uploaded notes**

To maintain order in the repository, every uploaded note is tagged according to subject, semester, or topic. This structured organization enables students to navigate easily and ensures that materials remain accessible to others in the correct academic context.

- **Search and filter functionalities for quick access**

An integrated search bar and filter mechanism help users quickly locate relevant notes based on keywords, uploader names, or subjects. This minimizes time spent searching for materials and enhances the platform's overall usability and efficiency.

- **Responsive user interface compatible across devices**

The frontend, built with React JS, is designed to automatically adjust to different screen sizes and resolutions. Whether accessed on a laptop, tablet, or mobile phone, users experience consistent navigation, fast rendering, and an intuitive interface.

3.1.2 Non-Functional Requirements

- **Scalability**

The system should be capable of handling increasing workloads as more users and files are added. By leveraging AWS EC2 auto-scaling and S3's elastic storage, the platform can dynamically expand or contract resources without affecting performance.

- **Security**

Ensuring user data protection is critical. All network communications occur over HTTPS, while AWS IAM policies control access between services. AWS Cognito handles secure authentication, and encryption is applied to data at rest in S3 and in transit through TLS.

- **Reliability**

The system must remain consistently available, even under heavy load or network interruptions. AWS's multi-region redundancy and automatic failover mechanisms ensure that user operations such as uploading or downloading are not disrupted.

- **Maintainability**

The application follows a modular architecture where each component (frontend, backend, and database) can be updated independently. This reduces downtime, simplifies debugging, and supports future upgrades like new features or design changes.

- **Usability**

The user interface emphasizes simplicity and clarity. Intuitive navigation, readable typography, and quick response times contribute to a smooth user experience, allowing first-time users to understand the platform's features without extensive guidance.

CHAPTER 4

4. DESIGN APPROACH AND DETAILS

4.1 System Architecture

The **NOTEFY** system follows a three-tier **cloud-based architecture**, designed to ensure modularity, scalability, and maintainability. The architecture consists of three main layers

- 1) the **frontend (presentation layer)**
- 2) **backend (application layer)**
- 3) **cloud infrastructure (data layer)**

each performing a specific role in the system's overall functionality.

The **frontend layer**, developed using **React JS**, forms the user interface through which students interact with the system. It manages navigation, data display, and input validation while providing a responsive and seamless experience across all devices. This layer communicates with the backend through RESTful APIs to fetch data, upload files, and manage tasks.

The **backend layer** is built using **Node.js and Express.js**. It acts as the logical core of the application, handling business rules, routing, and communication between the frontend and the cloud services. The backend also manages authentication, data validation, and API endpoints that interact with AWS services like S3 and DynamoDB.

The **cloud infrastructure layer** utilizes **Amazon Web Services (AWS)** for hosting, storage, and authentication. Files are stored in **AWS S3 buckets**, the backend is deployed on **AWS EC2 instances**, and **AWS Cognito** manages user authentication and access control. The data about users, uploaded notes, and tasks is stored in **AWS RDS** or **DynamoDB**, depending on the use case. This design ensures that all layers are loosely coupled, allowing independent scaling and maintenance.

4.2 Database Design

The database schema is designed to maintain simplicity while ensuring efficient data organization. The system primarily manages three types of entities — **Users**, **Notes**, and **Tasks** — each represented by a structured data model.

- **Users Table:** Contains user details such as username, email, and authentication tokens generated by AWS Cognito. Each user is uniquely identified by a User ID.
- **Notes Table:** Stores metadata of uploaded notes including file name, subject, uploader ID, upload date, and a reference URL pointing to the file's location in AWS S3.
- **Tasks Table:** Keeps records of the user's tasks with attributes such as task name, description, deadline, and status (pending or completed).

These tables are linked logically through the **User ID**, ensuring that all user-specific notes and tasks can be easily retrieved and managed. This relational or key-based design (depending on whether RDS or DynamoDB is used) provides fast access times and easy data scalability.

CHAPTER 5

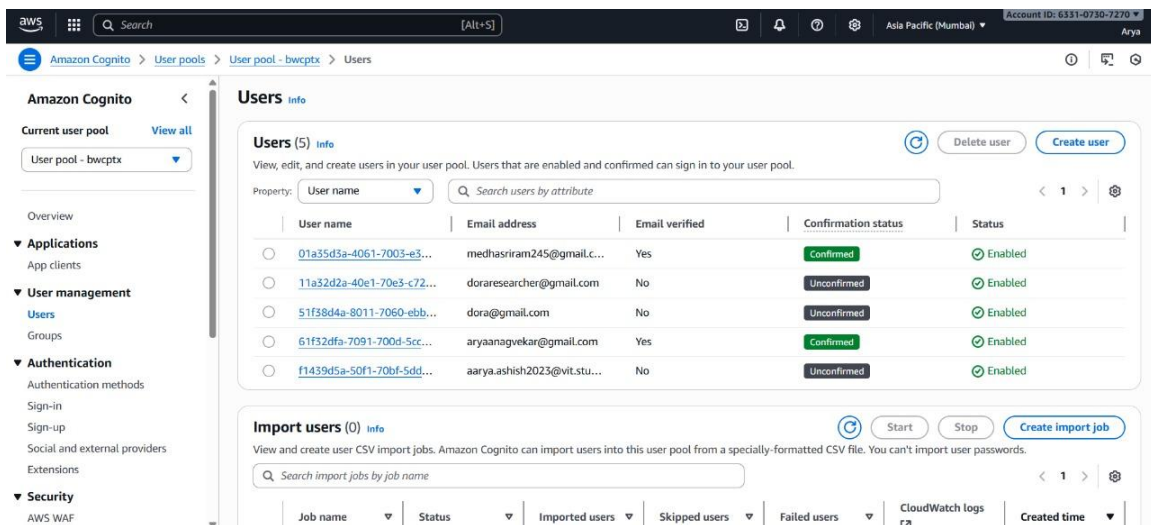
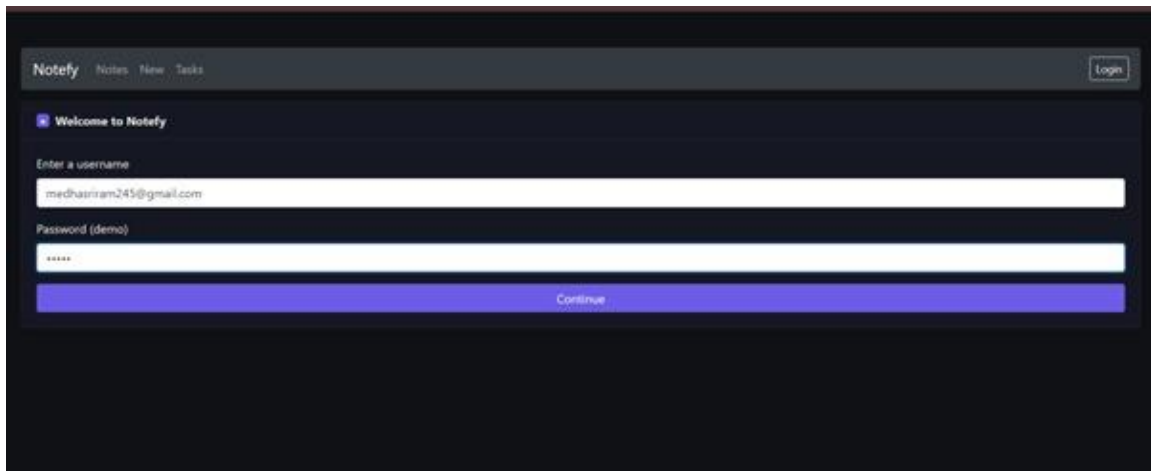
5. PROJECT DEMONSTRATION

The system interface includes:

- **Home Page:** Dashboard with navigation to Notes, Upload, and Tasks sections.
- **Notes Page:** Displays uploaded notes as categorized cards with title, description, and download options.
- **Upload Page:** Allows users to upload PDF or document files to AWS S3.
- **Todos Page:** Simple CRUD-based task management for personal use.

Screenshots of each module can be included here.

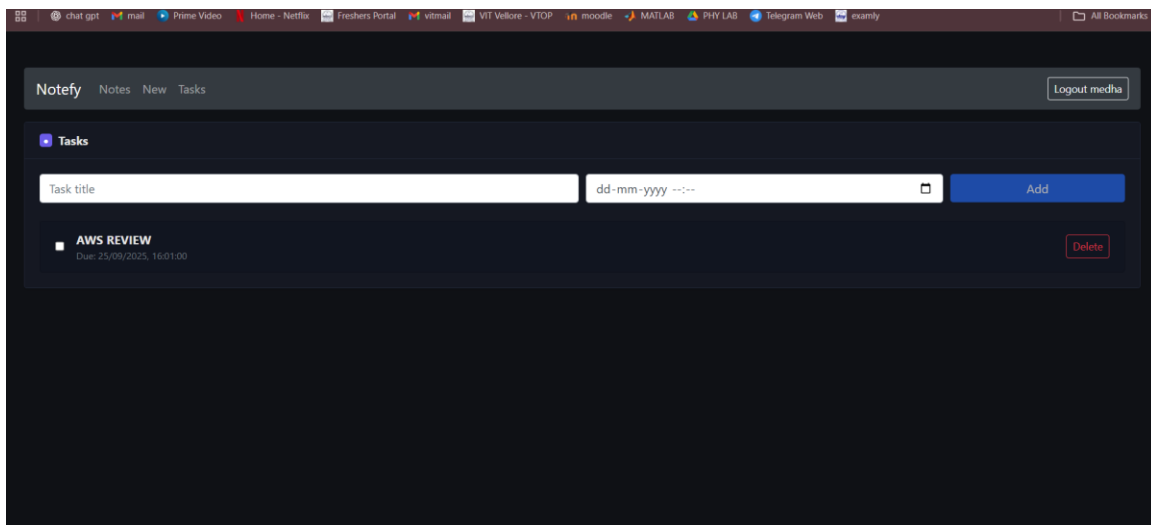
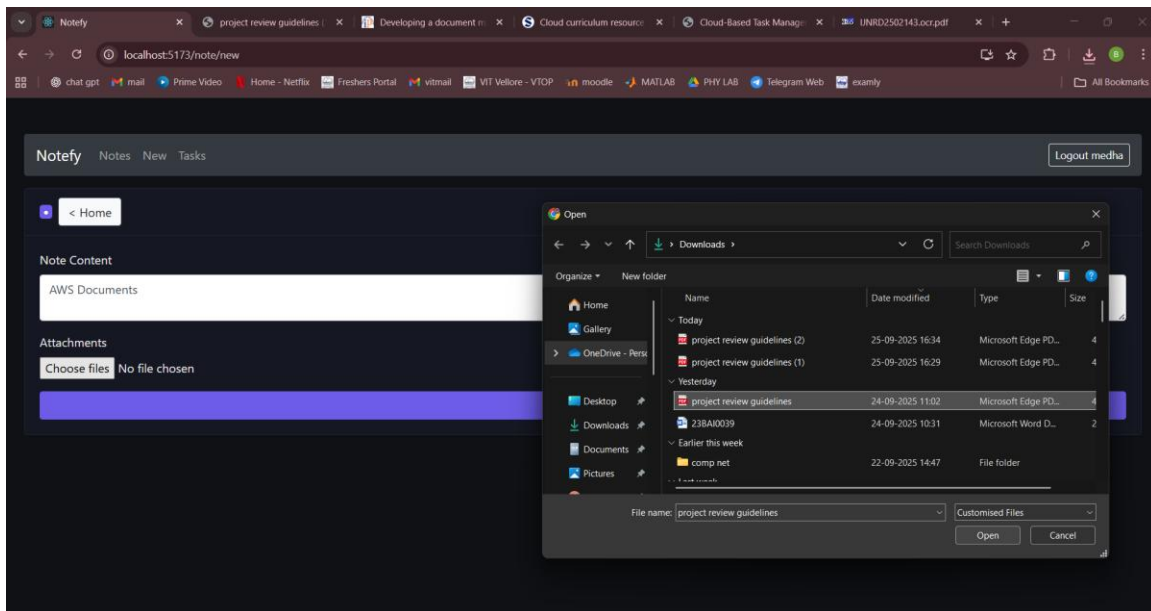
USER LOGIN – AWS COGNITO

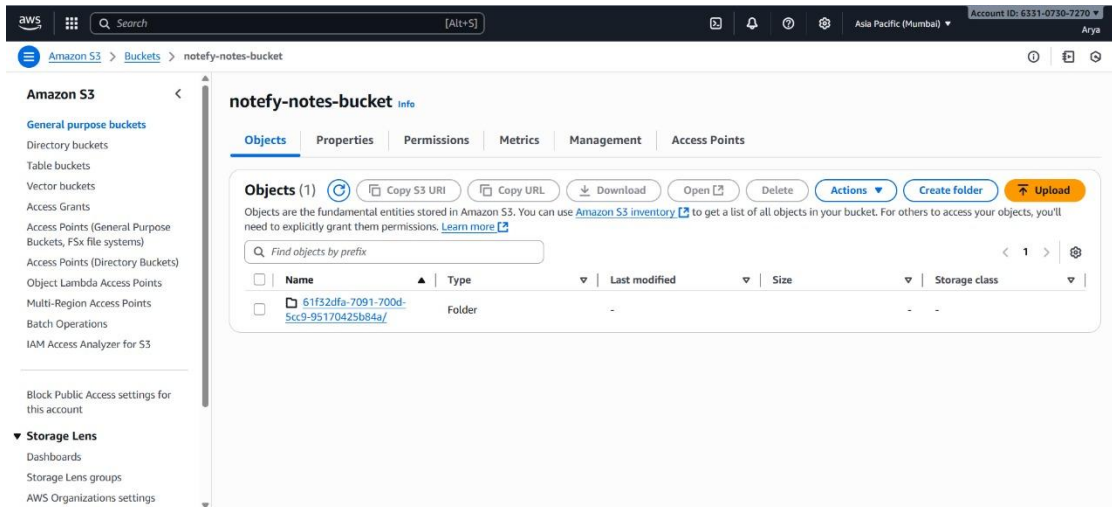


FUNCTIONS –

- Provides user management (sign-up/sign-in) functionality.
- Supports authentication and authorization for Notefy app.
- Users can be managed individually via the Users section
- Logging & Monitoring
- Status: Logging available

S3 bucket



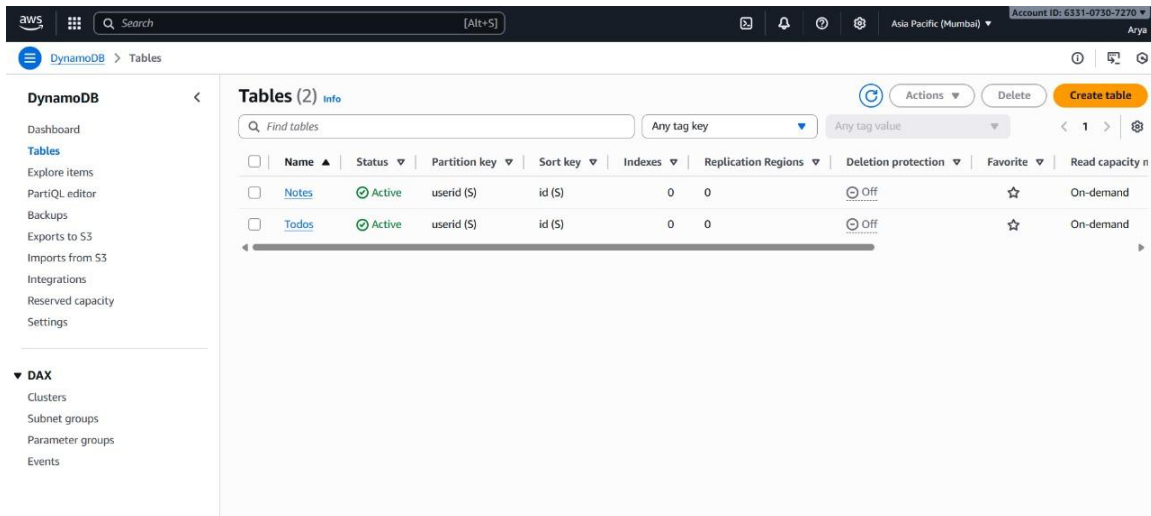


S3 Bucket Integration

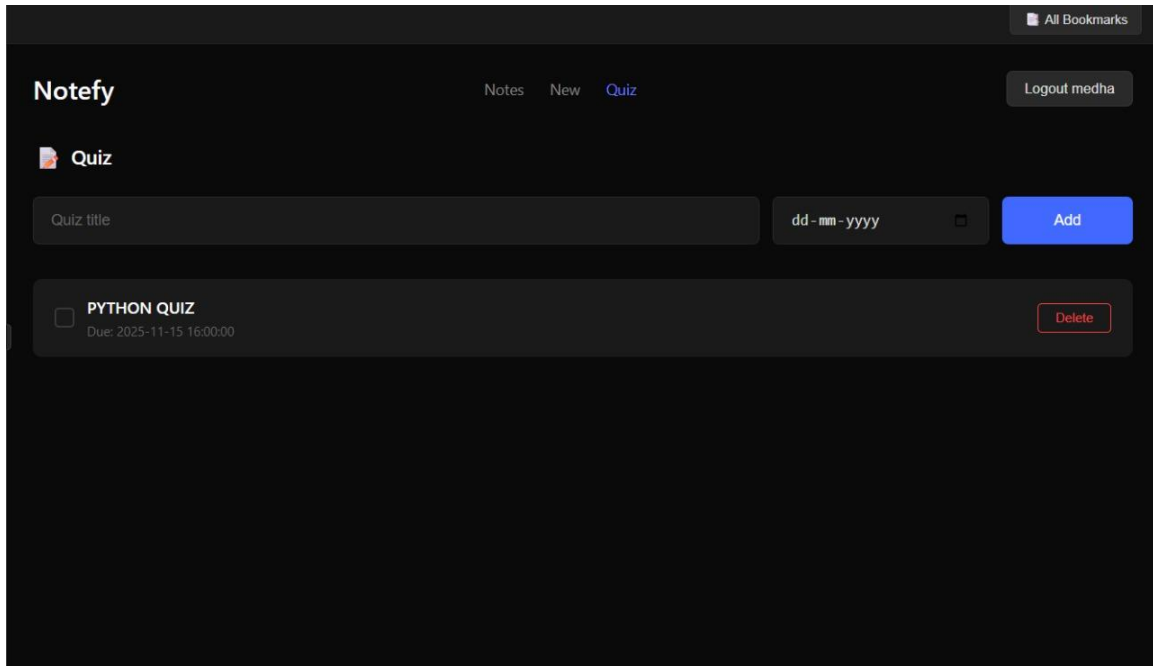
Details:

- Used to store uploaded notes and files from users.
- Configured with private access tied to Cognito authentication.
- Backend code uses AWS SDK to upload/download files.
- Supports memory-based upload via Multer in Express.

Dynamo DB Tables



PRACTICE WITH QUIZES:



Details:

- Stores note metadata, todos, and user info.
- Tables designed for fast read/write operations.
- Integrates with backend using AWS SDK.

CHAPTER 6

6. RESULT AND DISCUSSION

The project **NOTEFY – Student Notes Sharing and Task Management Website** was successfully designed, implemented, and deployed using AWS cloud infrastructure, fulfilling the objectives defined at the start of development. The system performs efficiently and demonstrates how cloud-based web applications can enhance academic collaboration and productivity among students. Through the combination of a responsive React-based frontend and a robust Node.js backend hosted on AWS EC2, the platform delivers fast and reliable user interactions, even when handling multiple requests simultaneously.

Testing confirmed that all major features such as user authentication through AWS Cognito, file storage and retrieval via AWS S3, and task management functionalities worked seamlessly. Uploading and downloading academic materials was fast and secure, with S3 providing efficient object storage and redundancy. CRUD operations for tasks were executed successfully, allowing students to create, update, and monitor their daily academic

goals in real time. The inclusion of search and filter functions further improved usability by enabling quick access to specific notes or resources.

Performance evaluations showed that the system maintained consistent response times and stable operation during concurrent usage. The integration of AWS services ensured scalability, fault tolerance, and data security throughout all operations. The website's interface was tested on multiple devices and browsers, confirming its responsiveness and accessibility. Students who reviewed the platform found it easy to navigate and appreciated the clean layout, smooth transitions, and minimal learning curve.

Overall, the results demonstrate that NOTEFY effectively meets its purpose as a unified digital platform for note sharing and task management. By leveraging the scalability and reliability of AWS, the project illustrates how cloud technologies can be applied to real-world educational systems. The combination of modern web development practices and cloud computing concepts resulted in a secure, efficient, and user-friendly application that aligns with current academic needs and future technological trends.

CHAPTER 7

7. CONCLUSION

The project NOTEFY – Student Notes Sharing and Task Management Website successfully demonstrates the integration of modern web technologies with cloud computing to create a practical and scalable academic platform. The system achieves its primary goal of simplifying how students share study materials and organize their academic tasks by combining two essential functions, note sharing and task management within a single, user-friendly interface.

Through the use of Amazon Web Services (AWS), the project effectively utilizes cloud resources such as EC2 for backend hosting, S3 for secure file storage, RDS or DynamoDB for data persistence, and Cognito for authentication. This configuration ensures that the system is both scalable and reliable, capable of supporting multiple users simultaneously without compromising performance. The modular architecture also allows for easy maintenance, expansion, and integration of additional features in the future.

From a development perspective, the project provided valuable insights into the practical implementation of full-stack cloud-based applications. It showcased the benefits of deploying services on cloud infrastructure, including reduced hardware dependency, improved accessibility, and automated scalability. Moreover, it demonstrated how web technologies like React and Node.js can work together efficiently to build dynamic and responsive user interfaces that enhance user engagement.

In conclusion, NOTEFY fulfills its vision of creating a centralized academic hub that promotes collaboration, productivity, and efficient resource sharing among students. Its success reinforces the growing role of cloud architecture in educational innovation, proving that scalable, cost-effective, and secure solutions can be built for real-world academic needs. In the future, the platform can be extended with advanced features such as AI-driven study recommendations, real-time chat for group collaboration, and mobile app integration — transforming it into a complete digital learning ecosystem.

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